

Two taxonomies for the building-block library

How each one is structured in the database, and how it was obtained

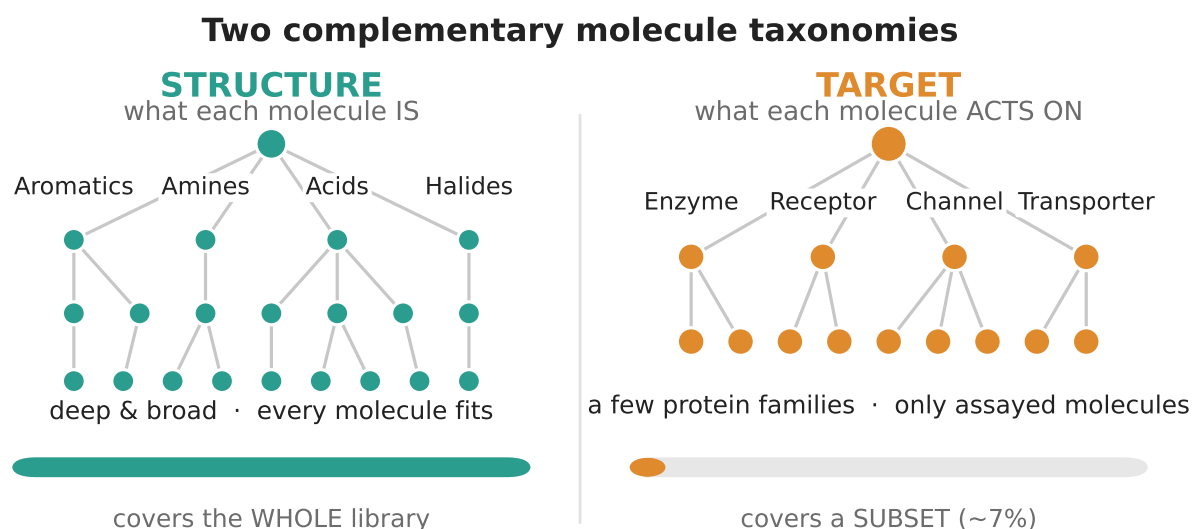
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Overview

Every commercial building block in our library (≈ 1.95 million unique molecules) is described along two independent axes, kept as **separate layers** so that neither constrains the other:

- a **structural** taxonomy — *what each molecule is* (its chemical skeleton and functional groups);
- a **target** taxonomy — *what each molecule acts on* (the protein families it has measured activity against).

Both are organised as **rooted trees** (a single top node; every class has exactly one parent), and both are stored in the same normalised shape: a *nodes* table (one row per class) plus an *edges* table (one row per parent→child link) — the standard adjacency-list representation of a hierarchy. Compound-to-class assignments live in *separate* fact tables, so the taxonomies themselves stay immutable and the assignment step is swappable.



1. Structural taxonomy — ChemOnt / ClassyFire

What it is. ChemOnt is the reference chemical-structure classification used by the ClassyFire system. It is a strict tree of **4,825 classes** under a single root (*Chemical entities*), **up to 11 levels** deep; the two top branches are *Organic* and *Inorganic* compounds, which split into 31 broad superclasses and then into progressively finer classes (e.g. *Benzenoids* → *Benzene derivatives* → *Benzonitriles*). It classifies structure only — not biological activity.

Database structure. Stored as an adjacency list: `chemont_nodes` (`chemont_id`, `name`, `definition`, `parent_id`, `depth`) and `chemont_edges` (`parent_id`, `child_id`); the full root-to-leaf lineage of every class is precomputed in `chemont_lineage.json`. Genuine per-molecule labels are held separately in `classyfire_ground_truth` (each row: molecule key, the kingdom/superclass/class/subclass/terminal it was assigned, the reconstructed lineage, provenance, and an evidence level).

How it was obtained. (i) We parsed the canonical `ChemOnt_2_1.obo` ontology file locally and verified its topology (single root, no multi-parent nodes, no cycles). (ii) We then retrieved **genuine** classifications for the molecules by exact structural-key (InChIKey) lookups against the authoritative ClassyFire services, over about two weeks of polite, automated querying, caching every response. (iii) Every returned class

identifier was validated against the local ontology before being accepted. This produced **200,001** genuine labels spanning 26 superclasses, 425 classes and 1,977 terminal classes; a fast local rule-based predictor was also benchmarked against them for the (larger) unlabelled remainder.

2. Target taxonomy — ChEMBL protein classes

What it is. A hierarchy of *protein target families* from the ChEMBL bioactivity database: a tree of **905 classes**, **6 levels** deep, under a single root (*Protein class*). Its top branches are broad functional groups — *Enzyme* (e.g. kinase, protease, cytochrome P450), *Membrane receptor* (e.g. GPCR families), *Ion channel*, *Transporter*, *Transcription factor*, *Epigenetic regulator* — each refined into specific families. It classifies *what a molecule binds*, not what it is.

Database structure. Same adjacency-list shape: `target_class_tree_nodes` (`class_id`, `parent_class_id`, `pref_name`, `class_level`, and the full path string) and `target_class_tree_edges` (`parent_class_id`, `child_class_id`). The molecule-to-class assignments are held separately in `target_class_layer_enriched` (each row: molecule, the protein target it was assayed against, and that target’s class path `class_l1 > class_l2 > class_l3`).

How it was obtained. The protein-family hierarchy is *not* exposed by the ChEMBL web API, so we downloaded the full **ChEMBL 37 SQLite release** locally and reconstructed the tree by joining four of its tables: `target_dictionary` → `target_components` → `component_class` → `protein_classification`. We then attached this tree to our molecules through their measured bioactivities. Coverage is intrinsically limited: only about **7%** of the library has any recorded ChEMBL activity, and of the targets involved $\approx 55\%$ carry a protein-family class (the rest are cell-lines, whole-organism or unchecked screening targets). This layer is therefore an *optional enrichment on a small, drug-like subset*, not a library-wide backbone.

At a glance

	Structural (ChemOnt)	Target (ChEMBL)
answers	what the molecule <i>is</i>	what it <i>acts on</i>
shape	rooted tree	rooted tree
size / depth	4,825 classes / 11 levels	905 classes / 6 levels
root	Chemical entities	Protein class
coverage	whole library ($\approx 100\%$)	a subset ($\approx 7\%$)
source	ChemOnt 2.1 + ClassyFire	ChEMBL 37 dump
stored as	<code>chemont_nodes/edges</code>	<code>target_class_tree_*</code>
role	canonical backbone	optional facet

Both trees, the per-molecule assignments, and all code are in the project repository (folder `database/v4/` for the trees, `reports/v4/` for documentation, `scripts/v4/` for the code). The two layers are deliberately independent: the structural taxonomy is the immutable backbone used everywhere, and the target taxonomy sits on top of it as an optional, partial enrichment.